FATHIMA AFRAH M A

AI Engineer | Software Engineer | Python | Generative AI | Back-End | SQL
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PROFESSIONAL SUMMARY

Computer Science graduate (B.Tech CSE, 2026) with strong foundations in data structures, algorithms, OOP, and software engineering. Builds and tests back-end software and AI applications in Python and C++, including RESTful APIs (FastAPI, Flask), SQL/SQLite data pipelines, Retrieval-Augmented Generation (RAG) applications with LLMs, and unit tests (GoogleTest, AddressSanitizer). Completed two remote internships, works independently with Git/GitHub, and has delivered multiple end-to-end projects. Working knowledge of Docker, CI/CD, AWS, and Microsoft Azure. Detail-oriented problem solver who writes clean, maintainable code.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript, HTML, CSS
Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Software Development Life Cycle (SDLC), Software Testing, Debugging, Agile
Back-End & APIs: Back-End Web Development, RESTful APIs, FastAPI, Flask, Server-Side Python, Streamlit
Databases: SQL, SQLite, MySQL
Testing & Quality: Unit Testing, GoogleTest, AddressSanitizer, Code Reviews
Tools, DevOps & Cloud: Git, GitHub, Docker, CI/CD, CMake, AWS, Microsoft Azure
Libraries & Data Analytics: Pandas, NumPy, Scikit-learn, OpenCV, Exploratory Data Analysis (EDA), Data Storytelling
AI & Machine Learning: Machine Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, LangChain, LangGraph, Prompt Engineering, Gemini API, TF-IDF, Logistic Regression

EXPERIENCE

Agentic AI Engineer Trainee (Internship) | Zetheta Algorithms Private Limited (Remote) **Oct 2026 – Present**
- Design, develop, and test Python applications and AI agent workflows; work independently using Git/GitHub.
- Building an Autonomous Financial Research Agent that automates research workflows using LLMs and APIs.

Artificial Intelligence Intern | CodeClause (Remote) **Jul 2025 – Aug 2025**
- Built Python pipelines for data extraction, preprocessing, and model evaluation, achieving 85%+ accuracy.
- Maintained SQLite data pipelines with data-quality checks; authored SOPs and technical documentation.

PROJECTS

ParcelPilot AI Support Copilot | Python, Streamlit, Gemini API, LLMs, RAG, AI Agents
- Developed an LLM-powered support investigation assistant in Python using a RAG pipeline with modular retrieval and response generation components.
- Integrated tools for order, account, and ticket lookup; combined policy and SOP documents with structured data to handle multi-step investigations.
- Showed supporting evidence for each decision; required explicit user confirmation before state-changing actions.

SmartVein-ID: Infrared Hand Vein Authentication System | Python, OpenCV, Raspberry Pi, SQLite
- Built a real-time, low-cost, contactless biometric authentication system on a Raspberry Pi Zero 2W using near-infrared imaging.
- Implemented an OpenCV pipeline (CLAHE, Black-Hat transform, Otsu thresholding, ROI extraction) with rotation-tolerant template matching, achieving 92–98% confidence scores; logged results to SQLite for traceability.

SecureBrowse: Privacy-Focused Web Request Engine | C++, libcurl, CMake, GoogleTest, Git
- Built a C++ web request engine that validates URLs, enforces TLS verification, blocks configured domains, and logs without exposing credentials.
- Wrote GoogleTest unit tests for invalid URLs, redirects, timeouts, and failures; used AddressSanitizer to find and fix memory errors.

MiniOS: Process Scheduling Simulator | C++, Operating Systems, Data Structures & Algorithms, Testing
- Implemented FCFS, Shortest Job First, and Round Robin scheduling algorithms and compared waiting, turnaround, and response times, with unit tests for edge cases.

Personality Prediction System | Python, NLP, TF-IDF, Logistic Regression, SQLite
- Built an NLP model classifying the Big Five personality traits from text using TF-IDF and tuned Logistic Regression, achieving 85%+ accuracy; evaluated with precision and recall.

EDUCATION

B.Tech – Computer Science & Engineering | B.S. Abdur Rahman Crescent Institute of S&T **2022 – 2026**
Higher Secondary (Class XII) | Oxford Matric Higher Secondary School, Thiruvarur **2020 – 2022**

CERTIFICATIONS

AWS Solutions Architecture Job Simulation, Amazon Web Services (Forage), Jun 2025; GenAI Powered Data Analytics Job Simulation, Tata Group (Forage), Jun 2025; Foundations: Data, Data, Everywhere, Google (Coursera), Jun 2025

Languages: English (Professional Proficiency), Tamil (Native)